

## Hospital Management System

1. Find alphabetical list of patients. If last name of patient is same order it according to first name and then middle name.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT firstName, middleName, lastName
FROM Patient1
ORDER BY lastName, firstName, middleName;
```

**Results** Explain Describe Saved SQL History

| FIRSTNAME | MIDDLENAME | LASTNAME |
|-----------|------------|----------|
| Meera     | Vishal     | Joshi    |
| Ayaan     | Ahmed      | Khan     |
| Rahul     | Rajendra   | Sharma   |

3 rows returned in 0.02 seconds [CSV Export](#)

2. Find all the patients who are treated under a particular consultant.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT firstName, middleName, lastName
FROM Patient1
ORDER BY lastName, firstName, middleName;

SELECT *
FROM Patient1
WHERE consultantId = 201;
```

**Results** Explain Describe Saved SQL History

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|---------|
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever   |

1 rows returned in 0.00 seconds [CSV Export](#)

3. Find all the patients who are associated with leading consultant 'A' and associate doctor 'B'.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT patientId
FROM PatientDoctor
WHERE doctorId = 201 AND role = 'Consultant'
INTERSECT
SELECT patientId
FROM PatientDoctor
WHERE doctorId = 203 AND role = 'Associate';
```

**Results** Explain Describe Saved SQL History

| PATIENTID |
|-----------|
| 103       |

1 rows returned in 0.00 seconds [CSV Export](#)

4. Find the list of patients admitted on particular date.

The screenshot shows the SQL Developer interface with the following SQL query:

```
)  
AND patientId IN (  
  SELECT patientId FROM PatientDoctor  
  WHERE doctorId = 203 AND role = 'Associate'  
);  
  
SELECT *  
FROM Patient1  
WHERE admitted_date = TO_DATE('2025-03-28', 'YYYY-MM-DD');
```

The results table displays the following data:

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|---------|
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever   |

1 rows returned in 0.00 seconds [CSV Export](#)

5. Find the tests undergone by specific patient.

The screenshot shows the SQL Developer interface with the following SQL query:

```
SELECT *  
FROM Patient1  
WHERE admitted_date = TO_DATE('2025-03-28', 'YYYY-MM-DD');  
  
SELECT testName, testDate  
FROM PatientTests  
WHERE patientId = 101;
```

The results table displays the following data:

| TESTNAME   | TESTDATE  |
|------------|-----------|
| Blood Test | 29-MAR-25 |
| X-Ray      | 30-MAR-25 |

2 rows returned in 0.00 seconds [CSV Export](#)

6. Find the list of all the patients who have undergone specific test.

The screenshot shows the SQL Developer interface with the following SQL query:

```
WHERE patientId = 101;  
  
SELECT DISTINCT *  
FROM Patient1  
INTERSECT  
SELECT DISTINCT Patient1.*  
FROM Patient1, PatientTests  
WHERE Patient1.patientId = PatientTests.patientId  
AND testName = 'ECG';
```

The results table displays the following data:

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM      |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|--------------|
| 103       | Ayaan     | Ahmed      | Khan     | 40  | Male   | 25-MAR-25     | 3      | 203          | Hypertension |

1 rows returned in 0.00 seconds [CSV Export](#)

7. Find the list of patients who have undergone at least two tests namely test1 and test2. (test1 and test2 are names of tests).

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
AND testName = 'ECG';  
  
SELECT patientId  
FROM PatientTests  
WHERE testName = 'Blood Test'  
INTERSECT  
SELECT patientId  
FROM PatientTests  
WHERE testName = 'X-Ray';
```

Results Explain Describe Saved SQL History

| PATIENTID |
|-----------|
| 101       |

1 rows returned in 0.00 seconds [CSV Export](#)

8. Find the list of patients who have undergone test1 or test2. (test1 and test2 are names of tests).

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT patientId  
FROM PatientTests  
WHERE testName = 'ECG'  
UNION  
SELECT DISTINCT patientId  
FROM PatientTests  
WHERE testName = 'Blood Test';
```

Results Explain Describe Saved SQL History

| PATIENTID |
|-----------|
| 101       |
| 103       |

2 rows returned in 0.01 seconds [CSV Export](#)

9. Find the list of patients who have undergone test1 but not test2. (test1 and test2 are names of tests).

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT patientId  
FROM PatientTests  
WHERE testName = 'Blood Test'  
MINUS  
SELECT DISTINCT patientId  
FROM PatientTests  
WHERE testName = 'ECG';
```

Results Explain Describe Saved SQL History

| PATIENTID |
|-----------|
| 101       |

1 rows returned in 0.00 seconds [CSV Export](#)

10. Find all the doctors who are 'Cardiologist'.

The screenshot shows a SQL Command window with the following query:

```
SELECT DISTINCT patientId
FROM PatientTests
WHERE testName = 'ECG';

SELECT *
FROM Doctor2
WHERE specialization = 'Cardiologist';
```

The results table shows one row:

| DOCTORID | NAME     | SPECIALIZATION | TYPE       | CONTACTNO  | EMAIL            |
|----------|----------|----------------|------------|------------|------------------|
| 203      | Dr. Khan | Cardiologist   | Consultant | 9876543211 | khan@example.com |

1 rows returned in 0.00 seconds [CSV Export](#)

11. Find all male/female patients. List the data alphabetically.

The screenshot shows a SQL Command window with the following query:

```
WHERE specialization = 'Cardiologist';

SELECT *
FROM Patient1
WHERE gender = 'Male'
ORDER BY lastName, firstName;
```

The results table shows two rows:

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM      |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|--------------|
| 103       | Ayaan     | Ahmed      | Khan     | 40  | Male   | 25-MAR-25     | 3      | 203          | Hypertension |
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever        |

2 rows returned in 0.00 seconds [CSV Export](#)

12. List the patients below age 12 and having cardiac problem.

The screenshot shows a SQL Command window with the following query:

```
SELECT *
FROM Patient1
WHERE gender = 'Male'
ORDER BY lastName, firstName;

SELECT *
FROM Patient1
WHERE age < 12 AND problem LIKE '%cardiac%';
```

The results section shows: no data found

13. List the patients who are referred by a given GP.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
WHERE age < 12 AND problem LIKE '%cardiac%';  
  
SELECT DISTINCT * FROM Patient1  
INTERSECT  
SELECT DISTINCT Patient1.*  
FROM Patient1, Referral  
WHERE Patient1.patientId = Referral.patientId  
AND gpName = 'Dr. X Gupta';
```

Results Explain Describe Saved SQL History

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|---------|
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever   |

1 rows returned in 0.00 seconds [CSV Export](#)

14. List the patients who are presently admitted in a given ward.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT DISTINCT Patient1.*  
FROM Patient1, Referral  
WHERE Patient1.patientId = Referral.patientId  
AND gpName = 'Dr. X Gupta';  
  
SELECT *  
FROM Patient1  
WHERE wardId = 1;
```

Results Explain Describe Saved SQL History

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM  |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|----------|
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever    |
| 104       | Meera     | Vishal     | Joshi    | 19  | Female | 26-MAR-25     | 1      | 204          | Fracture |

2 rows returned in 0.00 seconds [CSV Export](#)

15. List the patients who were admitted in some ward during given period between date1 and date2.

Home > SQL > SQL Commands

☒ Autocommit Display 10 Save Run

```
SELECT *  
FROM Patient1  
WHERE wardId = 1;  
  
SELECT *  
FROM Patient1  
WHERE admitted_date BETWEEN TO_DATE('2025-03-25', 'YYYY-MM-DD')  
AND TO_DATE('2025-03-28', 'YYYY-MM-DD');
```

Results Explain Describe Saved SQL History

| PATIENTID | FIRSTNAME | MIDDLENAME | LASTNAME | AGE | GENDER | ADMITTED_DATE | WARDID | CONSULTANTID | PROBLEM      |
|-----------|-----------|------------|----------|-----|--------|---------------|--------|--------------|--------------|
| 101       | Rahul     | Rajendra   | Sharma   | 25  | Male   | 28-MAR-25     | 1      | 201          | Fever        |
| 103       | Ayaan     | Ahmed      | Khan     | 40  | Male   | 25-MAR-25     | 3      | 203          | Hypertension |
| 104       | Meera     | Vishal     | Joshi    | 19  | Female | 26-MAR-25     | 1      | 204          | Fracture     |

3 rows returned in 0.00 seconds [CSV Export](#)